

RES. 6-012 Introduction to Probability - Lecture 19 & 20
Textbook Notes - Sections 5.4 & 9.1

• Section 5.4 Central Limit Theorem

• Architecture

$$\underbrace{S_n = X_1 + \dots + X_n}_{\text{sum of i.i.d.}} \longrightarrow \underbrace{Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}}_{\substack{\text{mean } 0 \\ \text{var } 1}} \longrightarrow \underbrace{Z_n \rightarrow N(0,1)}_{\substack{\text{5.4 CLT} \\ \text{De Moivre-Laplace} \\ \text{Binomial}}}$$

- 5.4 identifies the limiting distribution of a standardized sum is always Normal, regardless of the distribution of the individual terms.

• Statement

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance σ^2 .
Define:

$$Z_n = \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

By construction $E[Z_n] = 0$ and $\text{var}(Z_n) = 1$.

The CLT states:

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \Phi(z) \text{ for every } z$$

↳ where Φ is the standard Normal CDF.

- Scope:
- Requires only independence, identical distribution, and finite mean and variance.
 - The X_i may be discrete, continuous, or mixed.
 - No other assumption on their distribution.

Why $1/\sqrt{n}$ scaling?

- ↳ $M_n - \mu$ shrinks to zero (WLLN) and $S_n - n\mu$ grows without bound.
- ↳ Neither converges to anything informative.
- ↳ Dividing the deviation $S_n - n\mu$ by $\sigma\sqrt{n}$ is the unique scaling that holds the variance constant at 1, producing a non-degenerate limit.

• Normal Approximation Procedure

To approximate $P(S_n \leq c)$ where $S_n = X_1 + \dots + X_n$:

1. Compute $E[S_n] = n\mu$ and $\text{var}(S_n) = n\sigma^2$

2. Standardise: ~~z~~ $z = \frac{c - n\mu}{\sigma\sqrt{n}}$

3. Look up $P(S_n \leq c) \approx \Phi(z)$ in the normal tables

Equivalently: treat S_n as $N(n\mu, n\sigma^2)$

Accuracy: No general rule for how large n must be. Approximation is better when the X_i are symmetric (uniform X_i converge by $n \approx 8$; exponential X_i require far more), and better for c near the mean of S_n than in the tails.

• De Moivre - Laplace Approximation to the Binomial

For $S_n \sim \text{Binomial}(n, p)$, a sum of n i.i.d. Bernoulli (p) with $\mu = p$, $\sigma = \sqrt{p(1-p)}$, the CLT gives:

$$P(k \leq S_n \leq l) \approx \Phi\left(\frac{l + 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

• The $\pm 1/2$ continuity correction

- ↳ S_n is discrete, the normal is continuous
- ↳ Widening the interval by half a unit on each side captures the probability mass properly.
- ↳ Without it, $P(S_n = k)$ would approximate to zero.
- ↳ With it, single value probabilities become computable:

$$P(S_n = k) \approx \Phi\left(\frac{k + 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

- Quality:
 - Good for $n \geq 40-50$ when $p \approx 1/2$
 - Degrades as p approaches 0 or 1, requiring larger n .

• CLT vs. Chebyshev (Polling Revisited)

Both bound $P(|M_n - p| \geq \epsilon)$, but with very different tightness. Using the worst-case Bernoulli variance $p(1-p) \leq 1/4$:

Method	Requirement for $\epsilon = 0.01$, 95% confidence
Chebyshev (Section 5.2)	$n \geq 50,000$
CLT normal approximation	$n \geq 9,604$

CLT-based derivation

↳ By symmetry $P(|M_n - p| \geq \epsilon) \approx 2(1 - \Phi(2\epsilon\sqrt{n}))$

↳ Setting this ≤ 0.05 requires $\Phi(2\epsilon\sqrt{n}) \geq 0.975$,

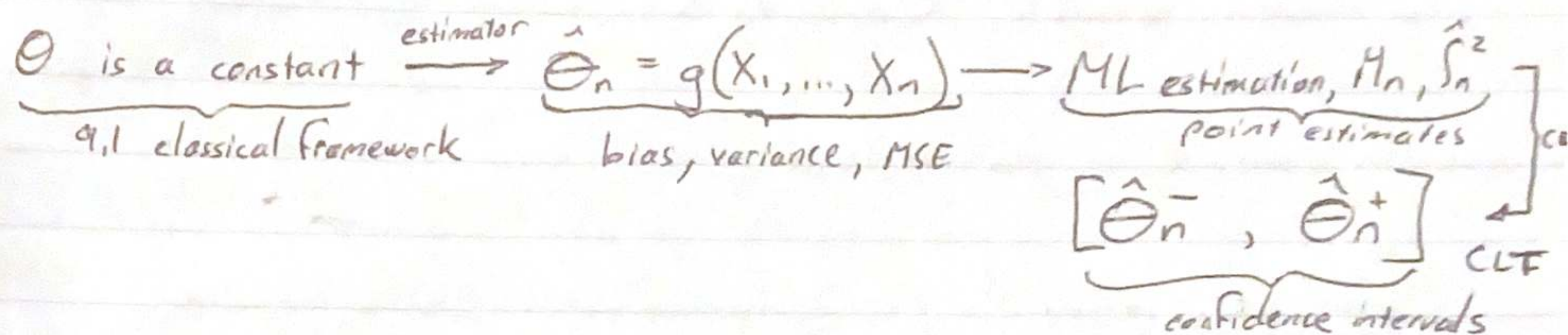
so $2\epsilon\sqrt{n} \geq 1.96$, giving $n \geq (1.96)^2 / (4\epsilon^2)$

→ The gap is the price of distribution-free guarantees:

↳ Chebyshev assumes nothing about shape, the CLT assumes approximate normality and gets a factor of five back.

• Section 9.1 Classical Parameter Estimation

• Architecture



Section 9.1 switches frameworks, θ becomes an unknown constant rather than a random variable, and uses the CLT from 5.4 to build confidence intervals around point estimates.

• The Classical Framework

θ is an unknown constant, not a random variable.

There is no prior (as in Bayesian Inference).

Instead of one ~~probabilistic~~ probabilistic model, there is a separate model $p_X(x; \theta)$ or $f_X(x; \theta)$ for each possible value of θ .

Notation: $\rightarrow$ the semicolon in $p_X(x; \theta)$ indicates functional dependence on θ (not conditioning).

$\rightarrow E_\theta[\cdot]$ and $\text{var}_\theta(\cdot)$ denote expectation and variance computed under the model corresponding to θ .

• Terminology for Estimators

Given observations $X = (X_1, \dots, X_n)$, an estimator is a random variable $\hat{\theta}_n = g(X)$.

An estimate is its realised value.

Estimation error :

$$\tilde{\theta}_n = \hat{\theta}_n - \theta$$

Bias :

$$b_\theta(\tilde{\theta}_n) = E_\theta[\tilde{\theta}_n] - \theta$$

Property

Unbiased

Asymptotically unbiased

Consistent

Definition

$$E_\theta[\tilde{\theta}_n] = 0 \text{ for every } \theta$$

$$\lim_{n \rightarrow \infty} E_\theta[\tilde{\theta}_n] = 0 \text{ for every } \theta$$

$$\hat{\theta}_n \rightarrow \theta \text{ in probability, for every } \theta$$

MSE decomposition :

$$E_\theta[\tilde{\theta}_n^2] = b_\theta^2(\tilde{\theta}_n) + \text{var}_\theta(\hat{\theta}_n)$$

This is $E[X^2] = (E[X])^2 + \text{var}(X)$ applied to $\tilde{\theta}_n$.

There is often a bias-variance trade-off : ~~reduce bias increases~~

↳ reducing variance frequently increases bias

A good estimator keeps both terms small.

• Maximum Likelihood Estimation

The ML estimate maximises the likelihood function over θ :

$$\hat{\theta}_{ML} = \arg \max_{\theta} p_X(x; \theta) \text{ or } \arg \max_{\theta} f_X(x; \theta)$$

For i.i.d. observations the likelihood factorises:

$$\prod_{i=1}^n p_{X_i}(x_i; \theta) \quad \text{or, more conveniently,}$$

$$\sum_{i=1}^n \log p_{X_i}(x_i; \theta)$$

• The log-likelihood has the same maximiser and converts to sums.

Interpretation: $\rightarrow p_X(x; \theta)$ is not the probability that the parameter equals θ .

$\rightarrow$ It is the probability of observing x when the parameter is θ .

$\rightarrow$ ML questions under which θ is the observed data most likely?

• Relation to MAP:

$\hookrightarrow$ ML is MAP with a flat prior as the prior term $f_{\theta}(\theta)$ becomes a constant and drops out of the maximisation.

$\hookrightarrow$ Every MAP calculation with a uniform prior is an ML estimation.

Invariance Principle: $\rightarrow$ If h is one-to-one, the ML estimate of $h(\theta)$ is $h(\hat{\theta}_{ML})$.

$\rightarrow$ Note: does not hold for LMS.

Asymptotics: $\rightarrow$ For i.i.d. observations under mild conditions, the ML estimator is consistent and asymptotically normal.

• Estimating the Mean and Variance

No distributional assumption required, these estimators work for any i.i.d. sequence with unknown mean θ and variation v .

Sample mean:

$$M_n = \frac{X_1 + \dots + X_n}{n}$$

↳ Unbiased ($E_\theta[M_n] = \theta$), $MSE = v/n$ independent of θ , and consistent by the WLLN.

Two variance estimators:

$$\bar{S}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - M_n)^2 \quad \hat{S}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - M_n)^2$$

Estimator	Bias	Note
$\bar{S}_n^2$	$E[\bar{S}_n^2] = \frac{n-1}{n}v$ biased, asymptotically unbiased	coincides with ML estimator when X_i are Normal.
$\hat{S}_n^2$	$E[\hat{S}_n^2] = v$ unbiased for all n .	the $n-1$ divisor corrects the bias

↳ For large n the two coincide.

↳ The $n-1$ arises because M_n is itself estimated from the data, consuming one degree of freedom.

Caution on "smallest variance":

~~Unbiasedness does not make an estimator good.~~

Unbiasedness does not make an estimator good.

The trivial estimator $\hat{\theta}_n = 0$ has zero variance but bias $-\theta$ and $MSE \theta^2$.

• Confidence Intervals

Fix a confidence level $1-\alpha$.

A $1-\alpha$ confidence is a pair of random variables.

$\hat{\Theta}_n^- \leq \hat{\Theta}_n^+$, both functions of the observations, satisfying:

$$P_\theta(\hat{\Theta}_n^- \leq \theta \leq \hat{\Theta}_n^+) \geq 1-\alpha \text{ for every possible value of } \theta$$

Interpretation (critical point):

↳ The randomness lies in the endpoints, not in θ .

↳ Once data is observed and interval evaluates to, e.g. $[-2.3; 4.1]$, no probability statement about θ is available (every qty is a constant).

→ Correct Reading: Repeating the procedure on indep. samples, roughly $1-\alpha$ of the intervals will contain the fixed θ , whatever θ happens to be.

• Constructing Confidence Intervals

→ Case 1: variance v known, X_i normal (exact)
 $M_n \sim N(\theta, \frac{v}{n})$ exactly:

$$\left[\hat{\Theta}_n - z \frac{\sqrt{v}}{\sqrt{n}} ; \hat{\Theta}_n + z \frac{\sqrt{v}}{\sqrt{n}} \right], \quad \Phi(z) = 1 - \frac{\alpha}{2}$$

For $\alpha = 0.05$, $z = 1.96$.

→ Case 2: variance unknown, n large (approx.)
Estimate v with $\hat{S}_n^2$ (invoke CLT):

$$\left[\hat{\Theta}_n - z \frac{\hat{S}_n}{\sqrt{n}} ; \hat{\Theta}_n + z \frac{\hat{S}_n}{\sqrt{n}} \right] \quad \rightarrow z \approx \text{approx in play}$$

↳ treat $\hat{\Theta}_n$ as Normal
↳ replace v by $\hat{S}_n^2$

→ Case 3: variance unknown, X_i normal, n small (t-dist)

↳ The statistic $T_n = \frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\hat{S}_n}$ is not Normal.

↳ For X_i , its dist does not depend on θ or V

↳ Use T distribution with $n-1$ degrees of freedom

$$\left[\hat{\theta} - z \frac{\hat{S}_n}{\sqrt{n}}, \hat{\theta} + z \frac{\hat{S}_n}{\sqrt{n}} \right], \quad \Psi_{n-1}(z) = 1 - \frac{\alpha}{2}$$

Properties of t distribution: (compare to Normal)

↳ Symmetric, bell-shaped

↳ More spread out with heavier tails

↳ At same ~~CL~~ CL, t-intervals wider than Normal.

↳ As n grows, t converges to Normal

↳ $n \geq 50$ Normal tables sufficient.

• Alternative Variance Approximations (Bernoulli case)

For i.i.d Bernoulli observations with unknown mean θ and variance $V = \theta(1-\theta)$, three options:

1. $\hat{S}_n^2$ → general-purpose unbiased estimator

2. $\hat{\theta}_n(1-\hat{\theta}_n)$ → plug estimate into known variance formula:
→ converges to V since $\hat{\theta}_n \rightarrow \theta$.

3. $1/4$ → conservative upper bound, since ~~CL~~
 $\theta(1-\theta) \leq 1/4$ for all $\theta \in [0,1]$

* For n in the hundreds or more, all three give nearly identical intervals.

* They diverge for small n .

Bayesian Inference vs. Classical Statistics

	Bayesian (Ch. 8)	Classical (Ch. 9)
Θ	random variable Θ with prior	unknown constant
Models	one joint model	one model per value of Θ
Output	posterior distribution	point estimate + conf interval
Point Estimates	MAP ; LMS ; Linear LMS	ML , sample mean
Interval Claim	$P(a \leq \Theta \leq b X=x) = 1-\alpha$ (credible interval)	Procedure covers Θ with prob. $1-\alpha$ (confidence interval)
Unverifiable Component	was the prior <u>reasonable</u> ?	did <u>this</u> instance fall in the α tail?